AWS Infrastructure Cost Estimation

Production Environment

Cargo Connect

AIT (AI Transformation) Platform

Gold Star Foods Group

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Executive Summary

This document provides a comprehensive AWS infrastructure cost estimation for the AIT (AI Transformation) Production environment at Gold Star Foods. The estimate is based on detailed workload assumptions and AWS Pricing Calculator outputs.

Cost Overview

Cost Category	Amount (USD)
Upfront Cost	$30.00
Monthly Cost	$4,302.57
Annual Cost (12 months)	$51,660.84

Top Cost Drivers

Service	Monthly Cost	% of Total
AWS Fargate (ECS)	$3,459.66	80.4%
AWS DMS	$240.39	5.6%
Amazon VPC	$226.40	5.3%
Aurora PostgreSQL	$226.10	5.3%
Amazon Route 53	$100.52	2.3%
Other Services	$49.50	1.1%

Infrastructure Assumptions

The following assumptions were used as inputs for the AWS Pricing Calculator to generate the cost estimates.

General Parameters

Parameter	Value
Database Storage	~150 GB
Monthly Work Hours	160 hours
React SPA Size	~100 MB
Production Monthly Deployments	~20
CloudFront Cache Misses	~900 MB

Database IOPS Profile

Metric	Value
Baseline IOPS (off-peak)	~20
Peak IOPS	~205
Burst IOPS (all 50 users active)	~400-500
Duration of Peak	8 hours/day

User Activity Assumptions

Metric	Value
Daily Active Users (DAU)	50
Average Usage per User	2 hours/day
Concurrent Active Users	5 per second
Actions per User per Second	~0.5 (1 action every 2 sec)
Total Actions/Second	~2.5

Note: Concurrent users calculated as 50 DAU × 2 hrs avg usage ÷ 8 hr workday × overlap factor ≈ 5 concurrent users during peak hours.

Monthly Data Volume Summary

Data Type	Monthly Volume
Disk I/O (Aurora)	~1.7 TB
Network (VPC internal)	~18 GB
Network (Cross-AZ)	~2-3 GB

Service-Specific Assumptions

AWS Secrets Manager

Secret	Access Pattern	Monthly API Calls
Aurora DB Credentials		
- ECS Task Startup (2 tasks)	~60 restarts/month	120
- Credential Rotation Check	None (manual)	0
DW2 SQL Server Credentials		
- DMS Instance	1 at startup + periodic	30
StockIQ API Credentials		
- Daily Sync Job	1/day	30
Subtotal		**180**
Buffer (caching, retries)	+20%	36
TOTAL API Calls		**~216**

AWS SSM Parameter Store

Parameter	Access Pattern	Monthly Interactions
OTEL Config		
- ECS Task Startup	2 tasks × 30 restarts	60
- OTEL Collector refresh	None (loaded at start)	0
Bastion SSH Key		
- Manual access (rare)	~5/month	5
TOTAL		**~65**

Amazon ECR (Container Registry)

Image Size Estimate

Component	Size
Base Image (Node.js Alpine)	~150 MB
App Dependencies (node_modules)	~200 MB
App Code	~50 MB
Total per Image	**~400 MB**

Storage & Data Transfer

Metric	Value
Images Retained	10
Total Storage	~4 GB/month
Inbound (CI/CD push)	~12 GB (30 deploys × 400 MB)
Outbound to ECS (same region)	FREE (private link)

Amazon CloudWatch

Custom Metrics (OTEL + App)

Metric Source	Metrics	Dimensions	Total
Bedrock (custom)	5	2 (model, endpoint)	10
Chat requests	2	2	4
HTTP requests	3	3	9
ECS Container Insights	10	2 (service, task)	20
ALB metrics	8	2	16
Aurora metrics	15	1	15
NAT Gateway	5	2	10
TOTAL			**~84**

Log Groups & Estimated Volume

Log Group	Source	Est. Volume/Day	Monthly
VPC Flow Logs	Network traffic	~500 MB	~15 GB
ECS Backend	App logs	~200 MB	~6 GB
ECS OTEL Collector	Telemetry	~50 MB	~1.5 GB
RDS PostgreSQL	DB logs	~100 MB	~3 GB
TOTAL			**~25.5 GB**

API Calls Summary

API Category	Monthly Calls
GetMetricData (Dashboard queries)	~20,000
PutMetricData (OTEL)	~8,640 (every 5 sec × 30 days)
Other (GetDashboard, ListMetrics)	~150
Logs Insights Queries	~50 GB scanned/month

Detailed Cost Breakdown

The following table provides a complete breakdown of all AWS services included in this estimate.

Service	Upfront	Monthly	Annual
AWS Fargate (ECS)	$0.00	$3,459.66	$41,515.92
AWS Database Migration Service	$0.00	$240.39	$2,884.68
Amazon VPC (NAT Gateway, IPv4)	$0.00	$226.40	$2,716.80
Aurora PostgreSQL Serverless	$0.00	$226.10	$2,713.20
Amazon Route 53	$0.00	$100.52	$1,206.24
Amazon CloudWatch	$0.00	$45.92	$551.04
Amazon CloudFront	$0.00	$1.91	$22.92
AWS Secrets Manager	$0.00	$1.20	$14.40
Amazon ECR	$0.00	$0.40	$4.80
AWS Systems Manager	$0.00	$0.05	$0.60
Amazon S3	$0.00	$0.02	$0.24
Amazon Cognito	$0.00	$0.00	$0.00
AWS Certificate Manager	$30.00	$0.00	$30.00
TOTAL	**$30.00**	**$4,302.57**	**$51,660.84**

Important Notes

⚠ **Disclaimer**

This document provides preliminary cost estimates based on projected usage patterns and is subject to change. Actual costs may vary depending on real-world usage, AWS pricing updates, and infrastructure adjustments.

Note: This estimate does not include Amazon Bedrock usage costs, which are provided in a separate document.

- All prices are in USD and based on US East (N. Virginia) region pricing.
- Fargate costs assume 2 tasks running 24 days/month with 4 GB memory each.
- DMS is configured for Multi-AZ deployment with t3.large instance.
- Aurora Serverless v2 with 4 ACUs (Aurora Capacity Units) on-demand.
- Route 53 cost is shared between Development and Production environments.
- ECR to ECS data transfer within the same region via private link is free.
- VPC Flow Logs (15 GB) qualifies for vended logs pricing tier.
- Cognito costs assume up to 1,000 monthly active users (MAU) within free tier.